

$$8 \cdot 8 \cdot 7 \cdot 5 = 64 \cdot 35$$

$$= 2240 \quad \checkmark$$

$$\begin{array}{r} 64 \cdot 35 \\ 35 \\ \hline 192 \\ 2240 \end{array}$$

2) $\underline{1} \quad \underline{1} \quad \dots \quad \underline{0} \quad \underline{0} \quad \underline{0}$ a) $2^5 = 32 \quad \checkmark$

b) $\underline{\quad} \quad \underline{\quad} \quad \dots \quad \underline{\quad} \quad \underline{\quad} \quad \underline{\quad} = 2^9 = 512 \quad \checkmark$

$\downarrow 1$ 2^9 $\downarrow 2$

c) 2^9 ways if num
odd $\Rightarrow 1$ else $= 0$
 $\Rightarrow 512$

d) 512 answers $\begin{matrix} 0 \text{ de } 1 - 1 \\ 2 \text{ de } 1 - \end{matrix}$ $P_3 = 3! = 6 \mid \Rightarrow 7 \text{ pos}$

e) $\underline{2} \quad \underline{1} \quad \underline{1} \quad \underline{1} = 2 \cdot 1 \cdot 1 \cdot 1 = 2 \quad \rightarrow 2 \text{ ntuning} \quad \checkmark$

3) 8 $\neq$ albe 1 albastru
 $1, 3, 5, 7$ $C_8^1 = 8 \text{ moduri}$

$$C_8^3 = \frac{8!}{3!5!} = \frac{6 \cdot 7 \cdot 8}{2 \cdot 3} = 56$$

$\Rightarrow 112 + 16 = 128$ $C_8^5 = \frac{8!}{5!3!} = 56$
 $C_8^7 = 8 \text{ moduri}$ $\checkmark$

4) $\{0, 1, 2, 3, 4, 5, 6, 7\}$

$$\frac{3}{3} \frac{4}{4} \frac{3}{3} \frac{3}{3} \frac{2}{2} = 3^3 \cdot 2^3 = 216$$

$$\frac{4}{4} \frac{4}{4} \frac{3}{3} \frac{3}{3} \frac{2}{2} = 4^2 \cdot 2^5 = 9 \cdot 32 = 288 \quad \Bigg| \Rightarrow \text{So } 4 \text{ moduli}$$

5) $A = \{1, 2, 3\}$ a) $f: A \rightarrow B$

$B = \{1, 2, 3, 4\}$ $4^3 = 64$

c) $g: B \rightarrow A$ b) $\begin{matrix} 1 \rightarrow 4 \\ 2 \rightarrow 3 \\ 3 \rightarrow 2 \end{matrix} \Bigg| \Rightarrow 4 \cdot 3 \cdot 2 = 24$

$3^4 = 81$

d) Since $|A| < |B| \Rightarrow$ at least one element from B will have the same image with another one.
 $\Rightarrow g$ is not injective

Also, f is not surjective because at least one element won't be the image of an element of A .

e) $\begin{matrix} 1 \rightarrow \\ 2 \rightarrow \\ 3 \rightarrow 2 \end{matrix} \Bigg| \Rightarrow 6 \text{ functions} \quad \begin{matrix} f(1) < f(2) < f(3) \\ | & & \downarrow \\ 1 & & 2 \end{matrix}$

6) $\{4, 5, 6, 7, 8, 9, 10\}$

$C_4^3 = 4$

a) $2^7 = 128 - 1 + 1 = 128$

$\downarrow \quad \downarrow$
 $\{ \emptyset \} \quad \{ 3 \}$

b) $\{1, 2, \dots, 10\} = \{ \quad ? \}$

$1 \rightarrow 9$ optiuni $3 \rightarrow 5$ optiuni $5 \rightarrow 1$ optiune
 $2 \rightarrow 7$ optiuni $1 \rightarrow 3$ optiuni $\Rightarrow$ Total = $9 \cdot 7 \cdot 5 \cdot 3$
 $= 63 \cdot 15$
 $= 945$

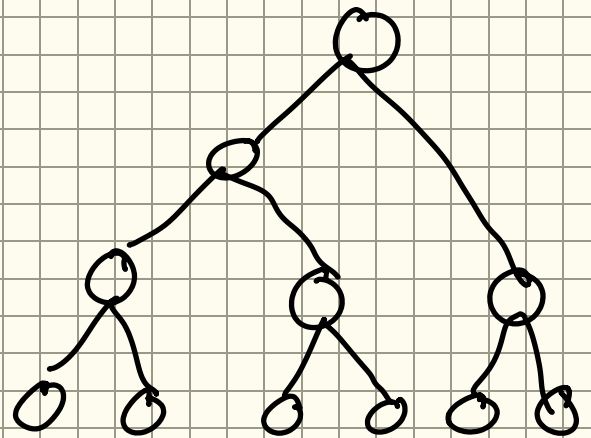
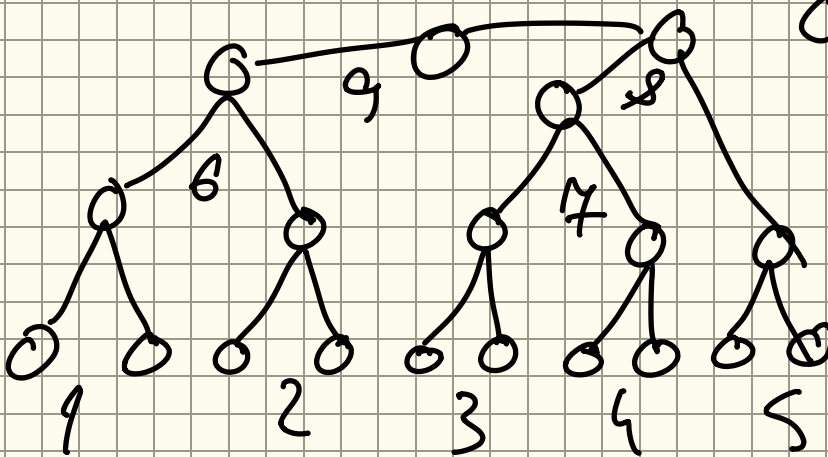
7) a) $64 \rightarrow 32$ meciuri
 $32 \rightarrow 16$ meciuri
 $16 \rightarrow 8$ meciuri
 $8 \rightarrow 4$ meciuri
 $4 \rightarrow 2$ meciuri
 $2 \rightarrow 1$ meci

$$1 + 2 + \dots + 32$$

$$1 + 2 + \dots + 2^5$$

$$1 \cdot \frac{2^6 - 1}{2 - 1} = 63$$

b) $100 \rightarrow 50$ de meciuri
 $50 \rightarrow 25$ de meciuri
 $25 \rightarrow 12$ meciuri

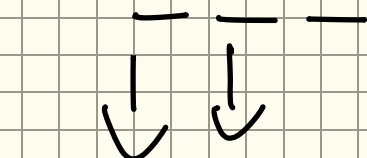


c) $n - 1$ meciuri

9
20

8)

3 Cai 3, 2, 1



3
 4000
 3100
 2110
 2200
 2111
 1111

???

$n:$
 $\begin{array}{r} 20 \\ \underline{11} \end{array}$

$\begin{array}{r} 300 \\ 210 \\ \underline{111} \end{array}$

9)

B F B F B F B F
F B F B F B F B

B
F B
B F
F B
2 B
3 F
4 2
5 1
6 8
7

? ? ? ?
' - - -

10) $M = \{1, 2, \dots, n\}$

$M = \{1, 2, 3\}$

$A = \{1, 1\}$

B